

Scaling Laws of Transient Spatiotemporal Error Bursts in Topological Quantum Error Correction: An Exploratory Study

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Abstract

Quantum computers built from many unreliable physical qubits can, in principle, be turned into a reliable machine by combining thousands of them into error-corrected “logical” qubits, but this guarantee assumes errors occur uniformly across the chip. Real hardware does not always behave this way: occasionally a cluster of qubits becomes briefly and sharply noisier than the rest, for reasons that are not always known. This paper asks a simple question about such events: can we predict how dangerous one is just by comparing its size, duration, and strength to the size of the code protecting against it, without first knowing what caused it?

More precisely, quantum error correction (QEC) guarantees that, below a critical physical error rate, increasing code size suppresses logical error arbitrarily. Real superconducting hardware violates the uniform-noise assumption behind this guarantee in a specific way: rare, localized, transient events, here termed “bursts,” temporarily elevate the error rate of a region of qubits. Google’s most recent below-threshold demonstration ([Google Quantum AI, 2025](#)) reports one such mechanism, affecting roughly 30 qubits for about 400 microseconds, that the authors themselves cannot explain. We do not attempt to explain its physical cause. Instead, we ask whether a burst’s danger can be predicted from a small number of dimensionless ratios: its spatial extent, duration, and strength, each measured relative to the code’s own scale, regardless of cause. We build a phenomenological and circuit-level simulation pipeline on top of Stim ([Gidney, 2021](#)) and PyMatching ([Higgott and Gidney, 2022, 2025](#)), run thirteen controlled experiments on a rotated surface code, and show that relative spatial scale is a strong, three-times-independently-validated predictor of single-burst damage; discover, and validate on out-of-family random shapes, that burst *orientation* relative to the code’s logical operator is an independent risk factor of comparable or greater magnitude that no prior model captures; and show, by testing against two independently-reported real hardware burst characterizations, that at real timescales a single burst event is essentially always catastrophic, reframing the open question from single-event severity to event frequency. We close by specifying the substantially different experimental design this reframing requires. All simulation code, experiment scripts, and raw results are available at <https://github.com/AlanPai777/qec-burst-scaling>.

1 Introduction

A single physical qubit is unreliable: it decoheres, is disturbed by its environment, and every operation performed on it carries a nonzero probability of error. Quantum error correction is the only known route from this reality to a computer capable of running algorithms of practical length. The approach encodes one *logical* qubit, the unit of information an algorithm manipulates, redundantly across many physical qubits, and repeatedly measures a set of indirect parity checks, the *syndrome*, that reveal whether an error occurred without disturbing the encoded information itself. A classical *decoder* reads the syndrome history and infers the most likely correction. The central theoretical guarantee, the threshold theorem, states that below a critical physical error rate (the *threshold*), increasing code distance d suppresses logical error arbitrarily. The *surface code* (Dennis et al., 2002) is the most hardware-compatible code known to admit this guarantee, and the full below-threshold scaling behavior has now been demonstrated experimentally, most recently on Google’s Willow processor (Google Quantum AI, 2023, 2025).

That same 2025 demonstration reports a phenomenon the threshold theorem does not anticipate. As the qubit array grew large enough, a new failure mode appeared: an anisotropic cluster of roughly 30 qubits would, approximately once per hour, become simultaneously unreliable, with the excess error decaying over roughly 400 microseconds. This is distinct from the previously characterized cosmic-ray relaxation mechanism (McEwen et al., 2022), which the Willow hardware’s gap-engineered design had already substantially suppressed. The authors state plainly that they do not know the cause, only that it sets a new, unexplained logical error floor near 10^{-10} . As spatially uniform noise is engineered down, rare, localized, transient noise bursts of unknown cause are emerging as the next bottleneck to fault-tolerant quantum computing, and neither the team that observed this phenomenon nor the broader literature reviewed below currently has a principled, cause-agnostic way to quantify how dangerous such an event is.

QEC theory almost universally assumes spatiotemporally uniform noise. A burst violates this assumption in a specific way: for some region, for some window of time, the error rate is far above background. As we detail below, existing analyses of burst-type noise assume the elevated-error region covers the *entire* code block, or model spatial correlation in a fully general way without isolating spatial extent as a controlled variable. The precise gap is: how does logical damage change as a burst’s spatial footprint grows from a single qubit to the whole code, relative to the code’s own size? This is not simply “more errors are worse,” which requires no study. The difficulty is that the same total excess physical error, distributed differently in space, time, or orientation, can produce very different consequences for a decoder that reasons about local syndrome patterns, and the code’s own protection is not spatially isotropic, since some directions of error propagation are geometrically cheaper to exploit than others. Quantifying a burst’s danger therefore requires characterizing the interaction between burst geometry and code geometry, an interaction no prior work, to our knowledge, has systematically measured.

We ask whether, for a localized burst with spatial radius r , duration τ , and peak strength p_b , injected into a rotated surface-code memory of distance d , background rate p_0 , and threshold p_{th} , the logical failure probability P_L collapses onto a low-dimensional function

$$P_L \approx F(\chi), \quad \chi = \left(\frac{r}{d}\right)^\alpha \left(\frac{\tau}{T}\right)^\beta g\left(\frac{p_b}{p_{\text{th}}}\right) \quad (1)$$

of dimensionless ratios that generalizes across burst geometries, or whether burst shape and temporal profile carry irreducible information no such description can capture. We test this by asking, first, whether under below-threshold background noise the logical damage of an isolated transient burst is controlled primarily by its relative spatial scale, relative temporal scale, and local strength, rather than by total physical error mass alone; and second, whether a

single dimensionless variable of the form above then predicts logical failure across code distance and, ideally, across burst geometry. Both questions concern a single, isolated burst. A third, related question, whether these conclusions still hold when bursts recur repeatedly over a long computation, is motivated by a specific tension we identify in the literature below, but sits outside this paper’s experimental scope; we return to it, as a fully specified future direction, at the end of this report.

Our contributions can be summarized as follows:

- We test, through thirteen controlled simulation experiments, whether relative burst geometry predicts logical damage, and find that relative spatial scale is a strong predictor, validated by three independent lines of evidence: consistency across every tested burst strength, a rigorous constrained-vs-unconstrained regression comparison, and reproduction under a genuine circuit-level noise model.
- We discover, through a shape-falsification experiment, that burst orientation relative to the code’s logical operator changes logical failure probability by a factor of $2.5\times$ to over $20\times$, growing with code distance, and we confirm, using randomly-generated, never-fit shapes, that this is a geometry-general phenomenon rather than an artifact of the specific shapes used to discover it.
- We test the resulting picture against two independently-reported real hardware burst parameter sets and show that, at real timescales, a single burst is essentially always catastrophic, which reframes the field’s open question from single-event severity to event *frequency*.
- We conduct a full-text literature review that identifies a precise, previously unstated gap between the two most relevant prior models of burst-type noise, and we specify a concrete, scoped experimental design for resolving the tension between them.

Before committing to this specific question, we deliberately enumerated several directions this line of work could plausibly serve, to make explicit which downstream claims a positive result would and would not support: a mechanism-agnostic risk descriptor, useful precisely because most observed correlated-error events currently have no identified physical cause; a safe-operating-region engineering criterion, generalizing an existing single-burst sustainable-threshold concept (Tan et al., 2026) from the whole-code limit to the finite-extent case; prioritization guidance for spatial versus temporal hardware mitigation investment, if the corresponding exponents differ substantially, which our results below show they do; a principled trigger criterion for adaptive, event-aware decoding, replacing the ad hoc, mechanism-specific thresholds used by current systems; a new hardware specification dimension complementing T_1 , T_2 , and gate fidelity; a translation layer between microscopic noise spectroscopy and code-level reliability, exercised directly in our final experiment; and a universality-class hypothesis, since a scaling collapse generalizing across burst shapes would be a novel application of finite-size scaling to a finite, transient perturbation rather than a uniform channel. Most of these directions are directly supported by the results below; a few depend on validation beyond this paper’s scope, and one higher-risk direction is deferred entirely to the closing discussion of future work.

2 Related Works

This section exists to answer one question before any experiment is run: has this problem already been solved? We trace, in turn, the hardware evidence that transient bursts are real, the theory that gives a code’s error threshold precise meaning, two directly competing models of what a

burst does to that threshold, the engineering systems already reacting to bursts in practice, and a handful of methodological tools we borrow from elsewhere, and we show that, taken together, they leave exactly one gap open, which the rest of this paper fills.

Before modeling “burst noise” abstractly, it is necessary to establish that it is a real, measured phenomenon, and to catalogue what is quantitatively known about it: a purely theoretical model with no anchoring in measured parameters risks answering a question nobody has. The experimental precedent for distance-dependent logical suppression on real hardware was established in 2023 ([Google Quantum AI, 2023](#)), which, read in full rather than at the abstract level, as was every paper cited in this section, also reports an early, quantitatively specific instance of a single burst distorting a scaling curve: in a distance-25 repetition-code experiment, a single high-energy event produced an apparent logical error floor of about 1.7×10^{-6} , and excluding samples near that event restored the expected exponential decay. This number directly informs the equal-error-mass experiment described below.

The foundational hardware evidence that a single ionizing-radiation impact can push a large fraction of a qubit array into simultaneous elevated error comes from a 26-qubit sub-array study ([McEwen et al., 2022](#)): an impact raised the number of simultaneously-decaying qubits from a baseline of about 4 to roughly 24 (nearly the entire monitored array) with an approximately 25-millisecond exponential recovery. This is direct hardware evidence that, on chips of that generation’s size, treating a burst as covering the entire code block, as both of the two most directly relevant prior models do ([Tan et al., 2026](#); [Sriram et al., 2025](#)), is a reasonable approximation, because the chip genuinely was small enough to be saturated by one event.

A synchronized-detector study using dedicated radiation detectors alongside a 10-transmon, $5 \times 5 \times 0.35 \text{ mm}^3$ chip reports a cosmic-ray-induced correlated-error rate of one event per 592^{+48}_{-41} seconds, accounting for only $17.1 \pm 1.3\%$ of all observed correlated-error events ([Harrington et al., 2024](#)). This single statistic is among the strongest motivations, in the entire review, for a mechanism-agnostic framing: even a team specifically instrumented to attribute correlated errors to cosmic rays cannot attribute most of them. An independent confirmation, using muon detectors and charge-parity monitoring on a $15 \times 15 \text{ mm}^2$ chip, reports quasiparticle relaxation constants of 2.4 ms at low injection power and 0.5 ms at high power ([Li et al., 2025](#)).

A chip already engineered to suppress the McEwen-type relaxation burst via superconducting-gap engineering was found to exhibit a new, residual mechanism nonetheless ([Kurilovich et al., 2026](#)): impact events produce a systematic negative qubit-frequency shift of up to 3 MHz, decaying over roughly 1 millisecond, at a rate of about once per 71 seconds, with, again, no identified physical cause. Together with Google’s own unexplained event, this establishes a pattern: the most carefully engineered current hardware continues to exhibit burst-type correlated errors of unknown cause, both the strongest motivation for a mechanism-agnostic analysis and the source of the two independent real parameter sets tested at the end of this paper’s experimental program.

The mapping between a topological code’s decoding problem and a disordered statistical-mechanics model, giving the accuracy threshold rigorous meaning as an order-disorder phase transition, is a foundational result ([Dennis et al., 2002](#)); a full-text search confirms this paper contains no discussion of localized or finite-region noise, supporting our novelty claim by elimination. The fitting methodology our scaling-collapse experiment draws on directly, locating a critical point by finding where logical-failure curves for different system sizes cross, then extracting the critical exponent via linear regression, with an explicit, checkable asymptotic bound on failure probability set by the number of homology classes of the code, was developed for a related setting ([Wang et al., 2003](#)). A different question, finite-temperature self-correction in higher-dimensional toric codes, supplies a methodological warning we carry into our discussion of future work ([Hastings et al., 2014](#)): a defect network becoming spatially connected, or

“percolating,” is not the same event as the code’s protection actually failing, and the two must be verified separately rather than conflated.

Given that burst-type events are real, does a surface code retain scalability if such an event occurs once, briefly? This is the most direct, tractable first question one can ask, and the one our first several experiments most directly extend. Answering it requires deciding how to characterize a burst’s spatial extent and duration, and how the decoder should treat a syndrome pattern shaped by a transient rather than a stationary process, decisions the uniform-noise theory above does not settle.

[Tan et al. \(2026\)](#), *Resilience of the surface code to error bursts*, is this paper’s single most important direct predecessor, and the paper whose abstract-level reading proved materially insufficient before we re-read it in full. Their model: a single, isolated elevation of the error rate, lasting exactly one syndrome-extraction round, applied uniformly across the entire code block, with the decoder always informed in advance of the burst’s timing. Using both a phenomenological and a full circuit-level depolarizing noise model, they define a burst threshold and a sustainable threshold and map their joint phase diagram, finding a surface code is resilient to even fairly strong single bursts if brief: under phenomenological noise, a burst rate of 7% depresses the sustainable threshold only from roughly 3% to roughly 2.7%. [Wang et al.’s \(2003\)](#) fitting procedure is adopted directly, and, under circuit-level noise specifically, the paper observes that some fraction of a burst’s errors are not detected until the following round, motivating our own decision to validate under genuine circuit-level noise rather than trust the phenomenological model unconditionally. Its stated limitation is the textual basis for this paper’s novelty claim: “It would be useful to extend our analysis to the case where errors in distinct rounds are correlated and error bursts have limited spatial extent, but we have not attempted that more general analysis” ([Tan et al., 2026](#)). This is not an inference we drew about the paper’s scope; it is the authors’ own statement.

A useful boundary case is the finding that coherent single-qubit rotation errors, not the Pauli/depolarizing-channel errors used above and by this paper, that are correlated and heavy-tailed can substantially degrade QEC protection ([Clader et al., 2021](#)). It establishes that “correlation and heavy tails are dangerous” is already known and therefore cannot be this paper’s novelty claim, but because its noise model differs in kind from ours, its specific numerical conclusions do not transfer; only its qualitative warning does.

[Tan et al. \(2026\)](#) answer how bad one burst is. [Sriram et al. \(2025\)](#), *Rare Events and Griffiths Phases in Topological Quantum Error Correction*, ask a substantially more alarming question: if bursts recur over the lifetime of a long computation, does the code retain a genuine asymptotic threshold at all? A real computation does not run for one round; a conclusion about an isolated event does not automatically extend to a system that must survive an unboundedly long sequence of rounds, some fraction of which may be elevated. Their setup lets each round of a long memory experiment independently have some probability of becoming a “rare round” with elevated, spatially-uniform error rate, with the observation window growing proportionally with system size, explicitly contrasted, in its own introduction, with the finite-duration, single-event setup described above ([Tan et al., 2026](#)). Their central argument is a largest-rare-region heuristic: the longest run of consecutive rare rounds in a long random sequence grows logarithmically with system size, and this single worst-case run dominates logical failure. For the one-dimensional repetition code, a new “Griffiths phase” appears in which failure degrades from exponential to stretched-exponential decay but the code remains decodable. For the two-dimensional surface code, this paper’s subject, there is no decodable Griffiths phase: once the rare-round error rate exceeds the bulk threshold, the code loses its threshold entirely in the thermodynamic limit, independent of how low the background rate is driven. A candidate explanation is further proposed, in a footnote, for Google’s own unexplained error floor: a non-monotonic dependence of failure on system size, minimized at a crossover length, the most direct existing connection in

the literature between a rigorous statistical-mechanics result and that open observation.

Two further papers refine methodological details without changing this conclusion. A warning that combined-statistic threshold fits can be biased by finite-size corrections at accessible code distances (Xiao et al., 2024) is carried into our scaling-collapse methodology, though its specific remedy is derived for biased-noise-tailored codes and does not directly transfer to our symmetric setting. A general tensor-network framework for arbitrary correlated noise, read in full, reports that a spatially-correlated cellular-automaton model can produce a complete reversal of expected distance scaling (Kam et al., 2026), a real, simulation-confirmed phenomenon we subsequently added to our own residual-analysis checklist.

The apparent contradiction between these two models resolves once they are seen as answering different questions: one about a single isolated event, one about indefinite recurrence. Critically, both burst models are spatially uniform across the entire code block; neither tests a spatially localized, finite-radius burst, under either framing. This is the most consequential gap this review identifies. This paper’s experimental program addresses the single-event half of that gap. The recurring-event half, whether finite spatial extent changes the recurring-burst conclusion, potentially reconciling the two models rather than leaving them in tension, is outside this paper’s scope and is specified as future work below.

Independent of whether a clean scaling law exists, practical systems already act on detected bursts. Q3DE, an architecture that detects anomalies via a statistical test on active-syndrome-bit counts and dynamically deepens the code, reduces affected duration roughly a thousandfold using measured hardware parameters (McEwen et al., 2022; Suzuki et al., 2022); its authors derive an explicit analytical model treating a burst’s affected size as an effective reduction in code distance, a competing baseline we include in our scaling-collapse comparison. A separate system targeting magic-state factories is important for narrowing rather than supporting this paper’s novelty claim (Chadwick et al., 2024): it already treats a finite burst radius as a controllable variable, but for the engineering problem of shutting down and remapping part of a factory, not the scaling-law question this paper asks about memory-experiment logical failure. The precise statement of this paper’s novelty is therefore not that finite-radius bursts have never been studied; that has already been done, for a different purpose (Chadwick et al., 2024); but that they have never been studied for surface-code memory threshold and scaling-law analysis. We revised our own framing to this more precise claim after reading this paper in full.

ReloQate, a system that predicts logical error rate from detector-firing rate and triggers a physical remap once a target logical-error-rate threshold is exceeded (Poster et al., 2026), is a concrete instance of the ad hoc, mechanism-specific threshold pattern a validated danger criterion would improve on. Promatch and the Local Clustering Decoder both address adaptive, real-time decoding for different problems than burst-danger assessment (Alavisamani et al., 2024; Zalawadiya et al., 2024): decoder-throughput scaling, and reaction to an already-known leakage-heralding signal, respectively. Two studies of leakage removal address a temporally-persistent but spatially gate-mediated, not radius-defined, process (Miao et al., 2023; McEwen et al., 2021), used here as temporal-axis rather than spatial-axis grounding. We searched specifically for a formal, burst-aware hardware specification standard extending T_1/T_2 /gate-fidelity reporting; none was found, only qualitative acknowledgment of the gap in adjacent benchmarking literature.

Several papers offer methodological tools adjacent to this project’s core question. Splitting-based rare-event Monte Carlo methods extend logical-failure-probability estimation as low as 10^{-20} (Bravyi and Vargo, 2013; Mayer et al., 2025), far below this paper’s target range of 10^{-1} to 10^{-4} , chosen deliberately for the reasons discussed in the methodology below; both require a custom Metropolis routine (the earlier paper reports roughly 20,000 CPU-hours), which we judge valuable for future extension but out of scope now. A Markov-chain Monte Carlo decoder approximating the true maximum-likelihood decoder, achieving a measurably higher threshold than minimum-

weight matching, has been proposed (Tobias et al., 2026); no public implementation was found. A semi-log survival plot is used to describe gate failure in hardware data, but, confirmed by full-text reading, does not employ formal survival-analysis statistics (Harper et al., 2025), an identified, unexploited cross-application opportunity. The single most directly actionable methodology paper found is *Dimensionally Consistent Learning with Buckingham Pi* (Bakarji et al., 2022), which formalizes the search for dimensionless groups as a constrained optimization problem over an exponent vector subject to dimensional-consistency constraints, giving a concrete, low-cost alternative to guessing scaling exponents one at a time. This directly motivates the methodology in our scaling-collapse experiment: rather than assuming a priori that radius and code distance combine as a single ratio, we let their exponents be fit independently and test whether the data support tying them together.

Taken together, this review rules out several claims that might otherwise sound novel: that correlated or heavy-tailed noise is dangerous; that rare regions can change scaling behavior; that a general correlated-noise framework can be built; that bursts can be detected and mitigated in real time; that rare failure probabilities can be efficiently estimated; and that finite spatial burst radius is, by itself, an unused concept: the magic-state-factory work above shows it is not, for a different purpose (Chadwick et al., 2024). What remains is precise: for a finite, localized, transient error burst, as opposed to one spatially uniform across the whole code block, is there a low-dimensional scaling variable, combining relative spatial scale, relative temporal scale, and local strength, that predicts logical damage to a surface-code memory, and that generalizes across burst geometries? This is the question our method and experiments address below. Its natural extension, whether the answer changes when such bursts recur, is specified, not answered, in the closing discussion of future work.

3 Research Design and Methodology

Given radius, duration, strength, and code distance, the most direct path to a high-accuracy predictive model is to feed all variables into a sufficiently flexible model and report its fit quality. We deliberately avoid this path for a specific epistemic reason: a flexible model fitting data well demonstrates only that flexible models can fit data, which is already known, and says nothing about whether the underlying phenomenon has a genuinely simple structure. The only way to learn whether a simple structure exists is to commit to a simple candidate description in advance and design experiments specifically capable of falsifying it. Machine learning is retained in this project’s design only as a diagnostic of last resort, to be used if and only if every simple model fails outright, a discipline adhered to throughout (no machine-learning model is fit at any point below).

This principle also governs our choice of noise model. Most of our experiments deliberately use a simplified, phenomenological-only noise channel, a single depolarizing error applied to data qubits once per round, so that spatial, temporal, and strength variables can be varied independently without confounding from gate- and measurement-level noise. Only after the core spatial-scaling result is established and rigorously fit do we introduce full circuit-level noise as an explicit validation step, checking whether the simplified model’s conclusion is an artifact of simplification rather than building the more realistic model from the outset.

Several analytical techniques from adjacent fields informed this design process, and we describe them here in the order we considered them, together with whether each already has precedent in the surveyed QEC literature or represents an unexploited opportunity. Comparing logical failure across different code sizes is, at its core, the same problem physicists face when studying a phase transition at different system sizes, so our first reflex was finite-size scaling and data collapse; the exact procedure for this setting exists already (Wang et al., 2003), which we adapt below

for a transient, localized perturbation rather than a uniform channel. A burst-weakened region growing large enough to connect one side of the code to the other suggests percolation theory, and a specific, relevant warning applies (Hastings et al., 2014): a defect network becoming spatially connected is not the same event as the code’s protection actually failing, which we do not resolve here but carry into our discussion of future work. Faced with several raw variables and wanting the smallest predictive combination of them, the natural reflex is dimensional analysis and the Buckingham π theorem; a concrete, adoptable algorithm for exactly this already exists (Bakarji et al., 2022), which we use directly in our scaling-collapse experiment below rather than guessing candidate exponents by hand.

A few further techniques were considered but not adopted, and we record why. A detector time series with a sharp rise followed by a slow decay, of a kind observed but not further analyzed elsewhere (McEwen et al., 2021), calls for wavelet or time-frequency analysis rather than a plain Fourier transform, since the signal is transient rather than stationary; we did not pursue this, but flag it as a natural way to refine our temporal-scale variable in later work. Rare, extreme tail probabilities call for extreme value theory, which no paper in our review applies to QEC logical-error tails specifically; we did not need it, since our experiments deliberately target a failure-rate range well above the extreme tail. Logical failure as a function of burst duration, with some trials never failing within the observation window, is a natural fit for survival analysis and hazard-rate framing in the style of Kaplan–Meier estimators; an adjacent framing without this formal machinery is used elsewhere (Harper et al., 2025), and we likewise did not adopt it, though it would likely use our fixed-shot samples more efficiently than the plain Bernoulli estimates we report. Two further reflexes we did put to direct use: held-out, out-of-family validation, standard statistical practice but not, to our knowledge, systematically applied to burst-shape generalization in the QEC literature we surveyed, is central to the shape-falsification and generalization experiments described below; and framing decoder performance with versus without prior knowledge of an event as a distribution-shift problem, which has direct precedent in real adaptive-decoding systems and in the single-event model discussed above (Tan et al., 2026), is reused directly in our decoder-awareness ablation.

All experiments simulate a rotated surface-code memory generated with Stim (Gidney, 2021), a high-performance stabilizer-circuit simulator, and decoded with PyMatching (Higgott and Gidney, 2022, 2025), a minimum-weight perfect-matching decoder that pairs triggered syndrome measurements on a weighted graph. This combination matches the toolchain used in the single-event model discussed above (Tan et al., 2026). Two noise models are used. The phenomenological-only model applies a single depolarizing channel to data qubits once per round, with no gate, measurement, or reset noise; the background rate was calibrated to $p_0 = 0.02$ after an initial value of 0.003 was found to produce zero observed logical failures at $d = 5, 7, 9$ within 20,000 to 50,000 samples, an uninformative baseline. The circuit-level model additionally enables gate, measurement, and reset noise, matching the circuit-level description used in that same model (Tan et al., 2026); its threshold, calibrated via curve-crossing, was found to be approximately 0.7%, consistent with commonly-cited literature values, typically 0.5% to 1%, for this noise model.

A burst is modeled as an additional depolarizing channel applied only to a selected qubit subset, only during a selected window of rounds. We built four spatial-selection primitives: a disk, the default, isotropic shape used for most of our experiments; a rotated ellipse, parametrized by semi-axes and orientation, used to test anisotropy while holding area fixed; a randomly-grown, irregular connected cluster of a specified qubit count, used as a genuinely out-of-family shape whose axes were never designed by the experimenter; and a principal-component-analysis routine that extracts the orientation and eccentricity of any selected qubit set, used to characterize the random clusters comparably to the hand-designed ellipses.

Injecting a burst initially located round boundaries by counting DEPolarize1 instructions, an

implicit assumption that background noise was the only source of that instruction type. This held under the phenomenological-only model but broke silently once circuit-level noise was introduced, since gate and measurement noise generate many additional such instructions per round. We resolved this by locating round boundaries via TICK instructions instead, empirically stable at exactly seven per round across all tested distances and both noise models, and confirmed by a regression test that this rewrite reproduces the phenomenological-only model’s results exactly. A further generalization allows a burst’s target probability to vary round by round, combined with an exponential-decay profile, enabling a burst that decays over time rather than only a flat-topped burst.

Each configuration is sampled between 8,000 and 100,000 times, and logical failure probability is reported with a 95% Wilson confidence interval rather than a bare point estimate; a configuration with zero observed failures is reported as an upper bound, never as $P_L = 0$. Two burst-strength thresholds were calibrated by curve-crossing: $p_{\text{th}} \approx 8.5\%$ for the phenomenological-only model, $\approx 0.7\%$ for the circuit-level model. For comparing candidate low-dimensional models, we consistently use leave-one-distance-out held-out validation: a candidate rescaling or model is fit using two of the three tested code distances and evaluated only on the held-out third. The question asked throughout the results below is never whether a model fits the data used to calibrate it, nearly always answerable in the affirmative for a sufficiently flexible model, but whether it correctly predicts data it has not seen.

4 Experimental Results

Every experiment below serves the same question stated at the outset: does a burst’s danger depend on its absolute size, duration, and strength, or only on these quantities relative to the code protecting against it? The thirteen experiments were not planned in full from the start; each experiment’s result determined the design of the next, in the order presented, and we flag at each step what the result changes about our answer to that question.

We first confirm the pipeline reproduces the most basic prediction of QEC theory. Under pure background depolarizing noise, no burst, at code distances 3, 5, 7, and 9, and after recalibrating p_0 to 0.02, logical failure probability decreased monotonically: 0.93%, 0.27%, 0.070%, and 0.040% respectively (Figure 1), with non-overlapping or only marginally overlapping 95% confidence intervals, the expected below-threshold behavior, clearing our pre-registered go/no-go criterion for proceeding further.

We next test the simplest, most easily falsifiable alternative to our central hypothesis: that a burst’s danger is determined solely by the total excess physical error it introduces, independent of how that quantity is spatially distributed. Two burst configurations at matched total excess error mass were compared: a compact-strong burst of radius 1.5 affecting one qubit at roughly 17% error rate, and a broad-weak burst of radius 3.5 affecting nine qubits at roughly 3.7%. The two configurations proved statistically indistinguishable at every tested distance (Figure 2; for example, at distance 5, 0.421% with confidence interval 0.383 to 0.463%, versus 0.397% with confidence interval 0.360 to 0.438%). Taken alone, this appears to support the mass-sufficiency hypothesis the experiment set out to falsify. The correct reading, however, requires noting that both tested radii are small relative to the code’s extent at all three distances; this null result holds entirely within a small-radius regime, and directly motivated the decision to push burst radius toward and past the code’s own scale in the next experiment.

With burst strength and duration fixed, radius was swept over the same absolute list of seven values at all three tested distances, so that relative radius varies naturally rather than being fixed by construction. Plotting logical failure against raw radius produces three visibly distinct curves; against relative radius, the curves collapse substantially closer together (Figure 3). The

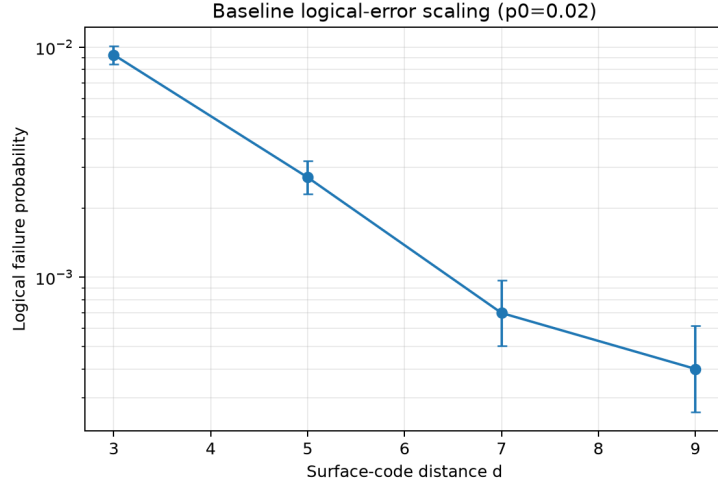


Figure 1: Logical failure probability versus code distance under pure background noise. Error bars are 95% Wilson confidence intervals.

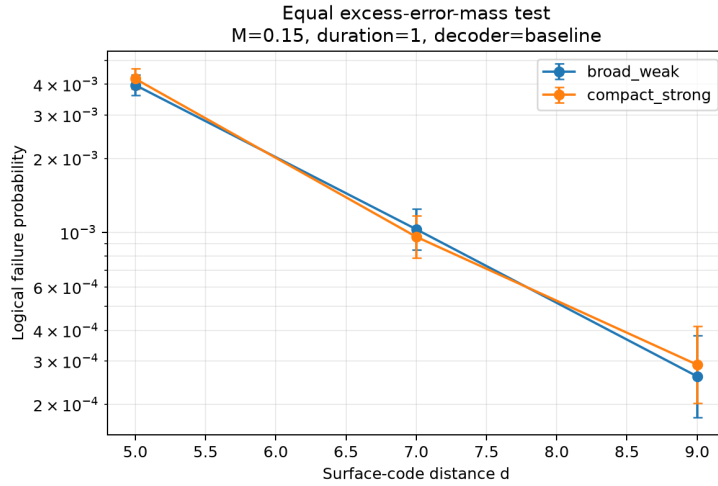


Figure 2: Logical failure probability for compact-strong versus broad-weak bursts of matched total excess error mass.

held-out prediction error drops from 0.71 for raw radius to 0.29 for relative radius, in log-scale root-mean-square error, the first strong, quantitative evidence that relative, not absolute, spatial scale controls burst damage.

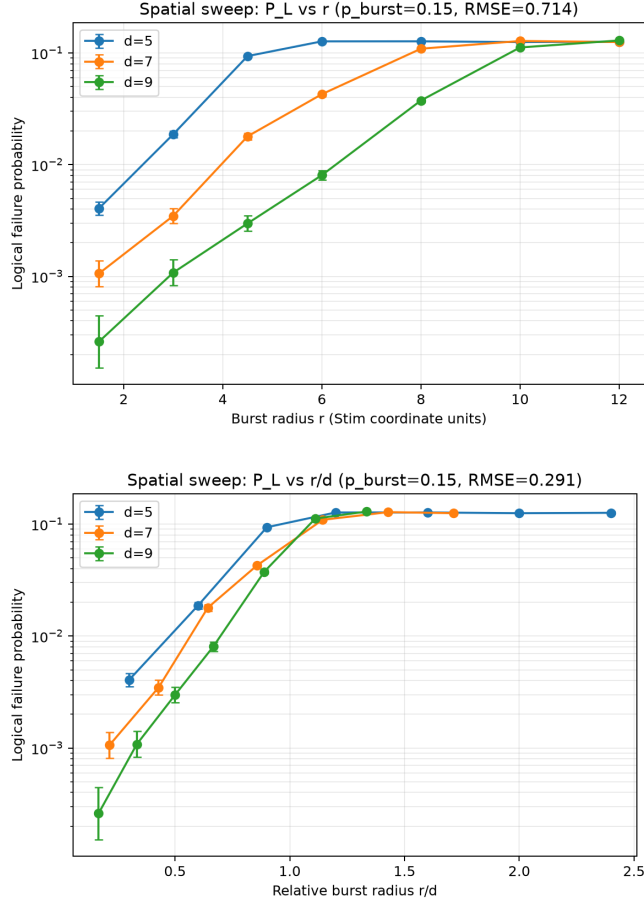


Figure 3: Logical failure probability versus raw burst radius (top) and relative radius (bottom), across three code distances.

With radius and duration fixed, strength was swept as a ratio to threshold from 0.5 to 8. No sharp kink appears at the threshold value itself (Figure 4); the increase is smooth in log-log scale, arguing against a simple binary above- or below-threshold danger criterion. Comparing the sub-threshold condition against the earlier no-burst baseline reveals a statistically significant increase of roughly two to two-and-a-half fold at distances 5 and 7, but no significant difference at distance 9, directly explained by the spatial-scaling result above, since the same absolute radius covers a smaller relative fraction of the larger code.

The full cross of the radius list and the strength ratios, 105 cells spanning seven radii, five ratios, and three distances, shows relative radius outperforming raw radius at every tested strength (Table 1), with the improvement growing from 21% at the weakest ratio to more than threefold at the strongest, hinting that the spatial exponent in a combined formula may not be strictly independent of strength, a detail carried forward below.

We then fit and rigorously test a candidate formula combining relative radius and strength ratio into a single dimensionless variable, following the methodology described above rather than assuming radius and distance combine as a single ratio by default. We first fit log failure probability as a function of log radius, log distance, and log strength ratio independently, three free parameters, testing whether the fitted coefficients on radius and distance are numerically opposite, which would justify treating them as a single ratio, rather than assuming it. We then

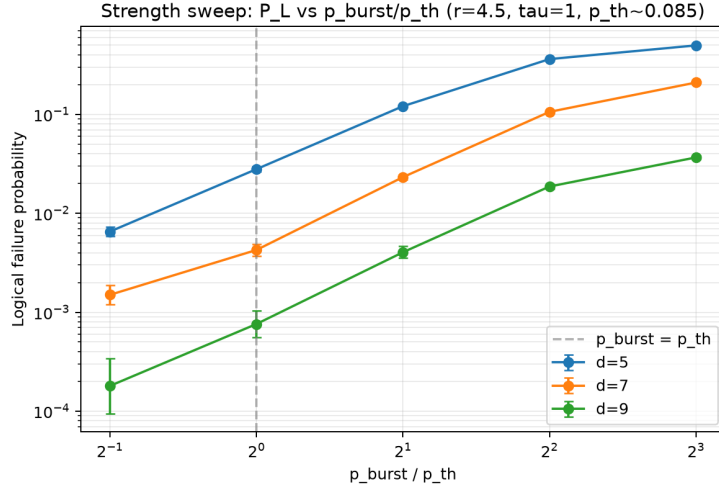


Figure 4: Logical failure probability versus burst strength relative to threshold. The dashed line marks the threshold value.

Table 1: Held-out RMSE, log scale, for raw versus relative radius, by strength ratio.

Ratio (burst strength / threshold)	RMSE, raw radius	RMSE, relative radius	Improvement
0.5	0.913	0.724	21%
1.0	0.865	0.506	41%
2.0	0.735	0.280	62%
4.0	0.768	0.316	59%
8.0	0.632	0.195	69%

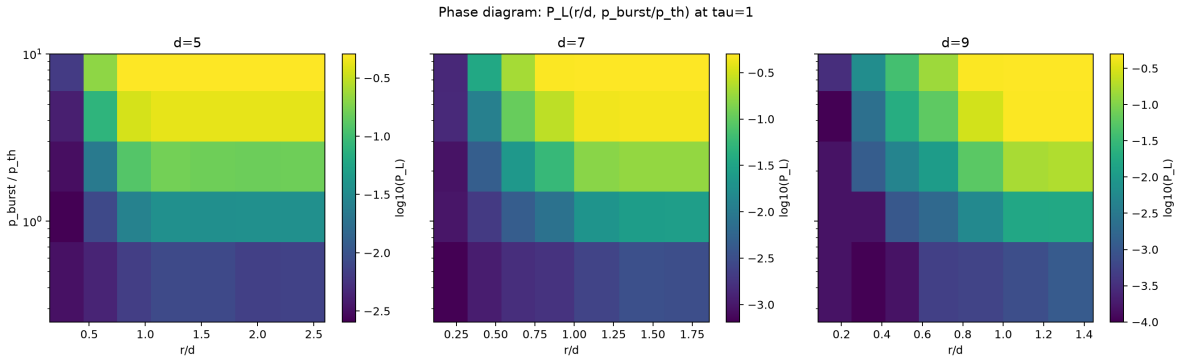


Figure 5: Logical failure probability as a function of relative radius and strength ratio, one panel per code distance.

compare this unconstrained model’s held-out accuracy against the constrained, two-parameter formula, and against three baselines: raw radius, total error mass, and an effective-distance formula proposed elsewhere (Suzuki et al., 2022).

The strict test rejects exact tying of radius and distance into a single ratio: the fitted coefficients are $+2.279$ with standard error 0.143 for radius and -3.454 with standard error 0.401 for distance, summing to -1.175 with standard error 0.425 , not consistent with zero at two standard errors. However, the held-out comparison (Table 2) tells a different story: the constrained, two-parameter formula achieves a held-out RMSE of 0.529 , matching, and marginally beating, the unconstrained three-parameter model’s 0.538 . The additional degree of freedom does not translate into better out-of-sample prediction. We interpret this as follows: the ratio of radius to distance is not an exactly precise physical law, but is the most parsimonious description that loses essentially nothing in predictive power relative to a strictly more flexible alternative. The fitted exponents were approximately 2.41 for the radius term and 1.61 for the strength term; the radius exponent substantially exceeding 1 hints that the relevant spatial quantity may be closer to burst area than to linear radius, a hypothesis we did not have time to test directly.

Table 2: Held-out RMSE, log scale, across candidate predictive models.

Model	Held-out RMSE
Constrained ratio-based formula (this work)	0.529
Radius and distance independent, unconstrained	0.538
Raw radius plus strength ratio, no distance normalization	0.727
Total error mass only	0.803
Effective-distance formula (Suzuki et al., 2022)	3.107

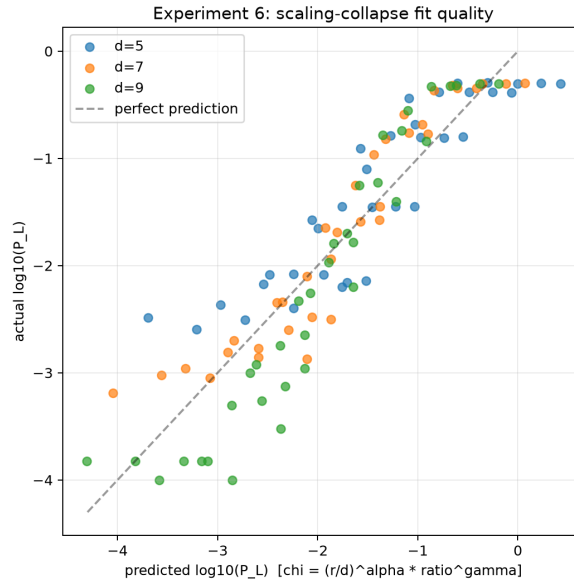


Figure 6: Predicted versus actual logical failure probability under the constrained formula, colored by code distance.

The effective-distance formula performs substantially worse (Table 2), but we emphasize this is not a fair indictment on its own terms: it was derived for a small-anomaly regime, and our test deliberately covers radii approaching and exceeding the code’s own size.

We next test whether the disk-fit formula correctly predicts damage for non-disk burst shapes of the same area. Rotated ellipses were constructed with semi-axes chosen so their area matched

a given target disk, at two eccentricities and two orientations, at fixed strength and across three distances. This produced this paper’s most consequential finding. Comparing two ellipses identical in every respect except a 90-degree rotation, and therefore containing the exact same number of affected qubits, eliminating any area or qubit-count confound, orientation alone changes logical failure probability by 2.5-fold to more than 20-fold, and this gap grows, rather than shrinks, with code distance (Table 3). The disk-only formula, blind to orientation, mispredicts these configurations by more than an order of magnitude in the most extreme case.

Table 3: Logical failure probability for identically-shaped, identically-sized bursts differing only in a 90-degree rotation.

Eccentricity	$d = 5$: 0° vs. 90°	$d = 7$: 0° vs. 90°	$d = 9$: 0° vs. 90°	Ratio at $d = 9$
Moderate	9.57% vs. 3.86%	3.42% vs. 0.68%	1.43% vs. 0.17%	$8.4\times$
High	2.06% vs. 0.57%	0.76% vs. 0.14%	0.32% vs. 0.03%	$10.5\times$

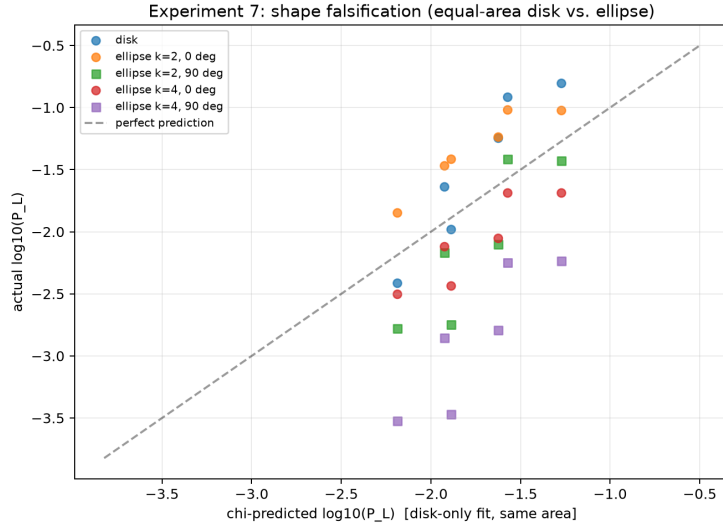


Figure 7: Predicted, disk-only formula, versus actual logical failure probability for disk and equal-area ellipse bursts; orientation is coded by marker.

To understand what this orientation dependence means physically, we inspected the compiled circuit’s logical observable and the coordinates of the qubits it references, finding the observable runs along a fixed coordinate in one direction while varying in the other: a straight line. The dangerous orientation is therefore perpendicular to the logical operator, and the safe orientation is parallel to it. This admits a percolation-style explanation consistent with a warning noted above (Hastings et al., 2014): flipping the tracked logical operator requires an error chain connecting a specific pair of code boundaries, and a burst elongated perpendicular to the operator is geometrically better positioned to help bridge that connection than one elongated parallel to it. A finer angle sweep across the full range from perpendicular to parallel shows the dependence is not a smooth trigonometric curve; failure probability remains high across roughly the first half of the range, drops sharply across the middle third, and flattens near the safe orientation (Figure 8), more consistent with a threshold-like transition. The transition’s shape is consistent between the two larger tested distances; the smallest shows some deviation, plausibly a discretization artifact, since intermediate angles do not preserve exact qubit count on the discrete simulation lattice.

The orientation effect above was discovered using clean, hand-designed ellipses; we next tested whether it generalizes to shapes never designed or used to calibrate anything. Connected, irregular clusters of exactly 21 qubits, matching a specific disk’s qubit count exactly and thereby

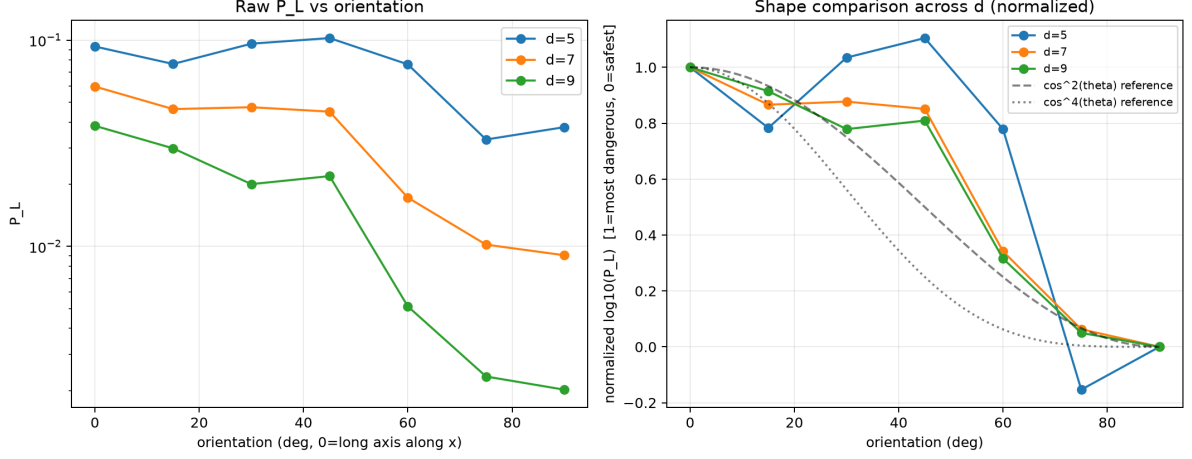


Figure 8: Logical failure probability versus burst orientation, normalized within each code distance so that 1 marks the most dangerous orientation and 0 the safest.

avoiding the area-count mismatch that complicated the cross-eccentricity comparison above, were grown via a randomized nearest-neighbor process from eight different random seeds, across three distances. Each cluster’s orientation and eccentricity were extracted by principal component analysis and combined into a single directional signal. The correlation between this signal and the disk-fit formula’s prediction residual was 0.516, correct in sign and moderate in strength, across 24 never-fit configurations (Figure 9), direct evidence that the orientation effect discovered on ellipses is not an artifact of that shape family, but a real, geometry-general phenomenon.

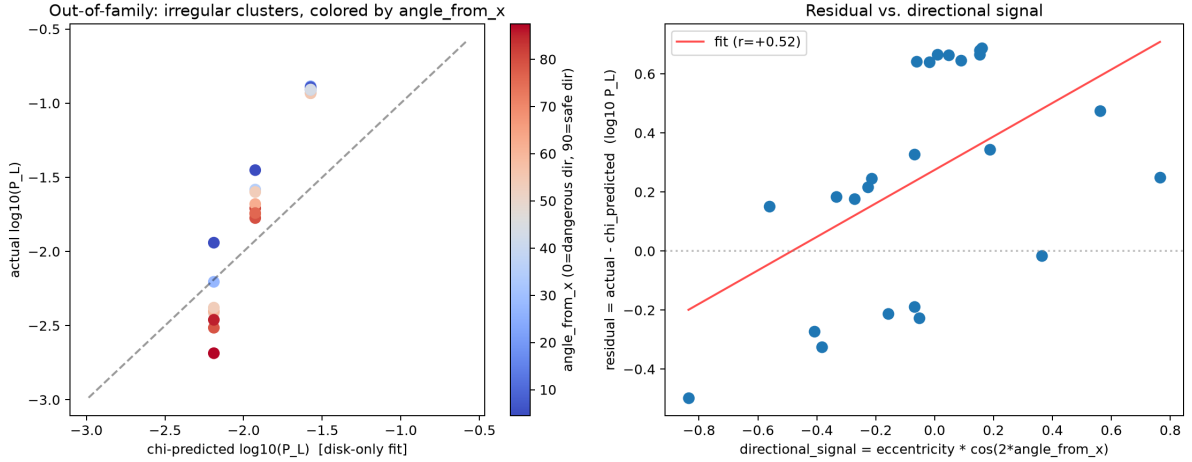


Figure 9: Left: predicted versus actual logical failure probability for randomly-generated irregular clusters, colored by orientation. Right: prediction residual versus the directional signal, with a linear fit.

The single-event model discussed above states an expectation that an informed decoder should perform nearly as well as an uninformed one, a claim established only in its whole-code-burst regime (Tan et al., 2026). We reran the 105-cell grid described above with the decoder informed of the true burst rather than built only from the background circuit, and compared directly against the existing data. Averaged across all configurations, the median gap between the two decoder conditions is close to zero, broadly consistent with the claim. Disaggregating by strength and by relative radius, however, shows the claim does not generalize uniformly (Table 4, reporting the mean gap within each stratum): the informed decoder’s advantage grows to roughly one-and-a-half fold at the highest tested strength, and is largest specifically when the burst is small relative to the code, vanishing only near whole-code coverage, the only regime

actually tested previously (Tan et al., 2026). Because Google’s real event is precisely a small, strong, finite-radius burst, this result suggests that for the bursts most representative of current hardware, decoder awareness is a measurable, actionable mitigation rather than a negligible refinement.

Table 4: Mean decoder-awareness gap, difference in log failure probability between an uninformed and an informed decoder, by strength and by relative radius.

By strength ratio	Gap	By relative radius	Gap
0.5	−0.039, not significant	Small (0.16–0.73)	+0.128
1.0	−0.029, not significant	0.73–1.28	+0.008
2.0	+0.062	1.28–1.84	+0.019
4.0	+0.127	Near whole-code (1.84–2.40)	−0.008
8.0	+0.181, roughly 1.5×		

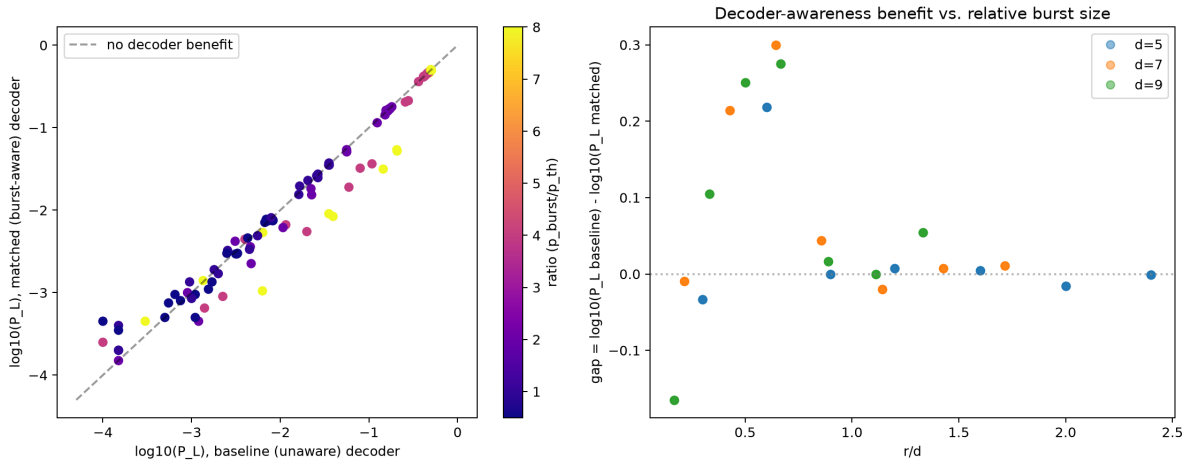


Figure 10: Left: matched- versus baseline-decoder failure probability, colored by strength. Right: gap versus relative burst radius.

The experiments above all use the simplified phenomenological-only noise model. We repeated the spatial-scaling sweep under genuine circuit-level noise, requiring the burst-injection rewrite described in the methodology, and validated by a regression test confirming the phenomenological path’s exact numbers were reproduced. The circuit-level threshold, independently calibrated at approximately 0.7%, matches the commonly-cited literature range, a useful correctness check. Rerunning the spatial sweep reproduces the phenomenological result closely: the relative-radius held-out RMSE is 0.285, essentially identical to the phenomenological value of 0.29, and still far better than raw radius, 0.53 versus 0.71 phenomenological (Figure 11). The finding that relative spatial scale controls single-burst damage is not an artifact of the simplified noise model.

Finally, we test this framework against real, independently-reported burst parameters: Google’s unexplained event, roughly 30 qubits, anisotropic, roughly 400-round exponential decay, using an approximate one-microsecond-per-round conversion whose uncertainty we state explicitly; and a residual phase-error burst reported elsewhere (Kurilovich et al., 2026), roughly 1000-round decay, with no spatial extent reported in that source, so a disk of a stated, non-literature-derived size was substituted. Neither source reports a value directly convertible to our simulation’s error probability, so strength was swept rather than asserted. This required implementing round-varying burst strength, since real bursts decay rather than remaining flat-topped, tested at distances 5 and 9.

Nearly every configuration saturated to a failure probability near 50%, the ceiling corresponding

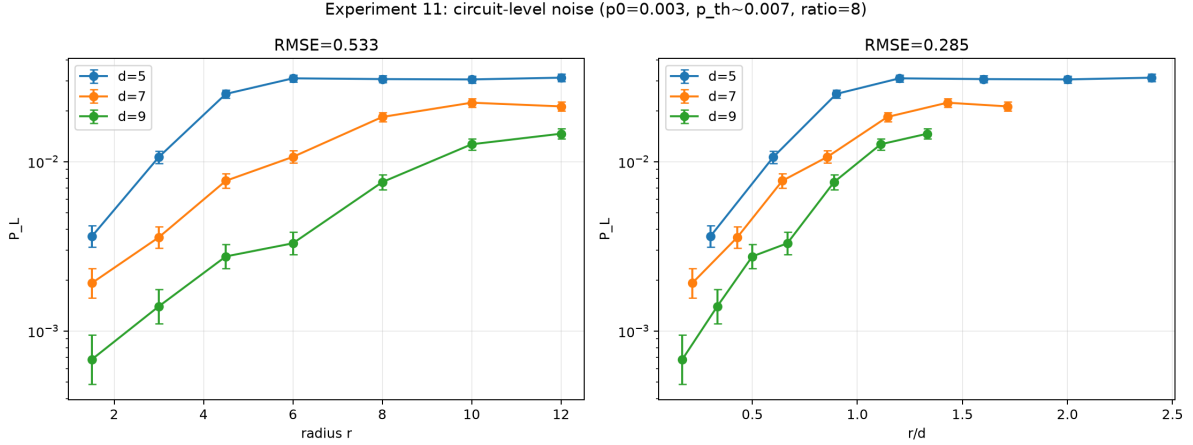


Figure 11: Logical failure probability versus raw radius (left) and relative radius (right) under full circuit-level noise.

to a decoder that has effectively lost all information (Table 5). The sole exceptions are the larger tested distance at the weakest tested strength, peak strength exactly at threshold rather than above it, giving 30.5% for the Google-inspired scenario and 34.4% for the Kurilovich-inspired scenario, still far higher than anything observed in the short-duration bursts earlier in this program. We deliberately do not attempt a numerical comparison against a temporal-axis fit for these two non-saturated points: their relative duration is roughly 14 to 35 times beyond the maximum tested in the temporal-scaling experiment above, and extrapolating that far would not be a credible claim.

Table 5: Logical failure probability for physically-inspired burst scenarios. Values near 50% indicate saturation.

Scenario	Distance	Qubits affected	Ratio 1	Ratio 2	Ratio 4
Google-inspired, anisotropic	5	15	49.5%	49.7%	50.1%
Google-inspired	9	27	30.5%	49.5%	50.4%
Kurilovich-inspired, disk	5	25	49.9%	49.9%	50.1%
Kurilovich-inspired	9	29	34.4%	50.1%	50.2%

At real hardware timescales, a burst matching either independently-reported real event is essentially always catastrophic if it coincides with an active memory experiment, consistent with the source papers’ own language of catastrophic bursts and error floors. This reframes the central open question this framework can address: not how severe a single event is, essentially answered above as almost always maximally severe, but how frequently such an event occurs relative to the length of a computation, a question the single-isolated-event framing used throughout this program cannot, by construction, answer, and which we specify as future work below.

5 Discussion

A useful standard for judging a piece of research is not whether it proves a theorem or reports a positive result, but whether it changes what an informed reader would have believed beforehand. Before this project, a reasonable prior might have been that burst size, duration, and strength are roughly comparably important, with a single event’s severity roughly proportional to the number of errors it introduces. Our results overturn that belief twice: spatial scale matters

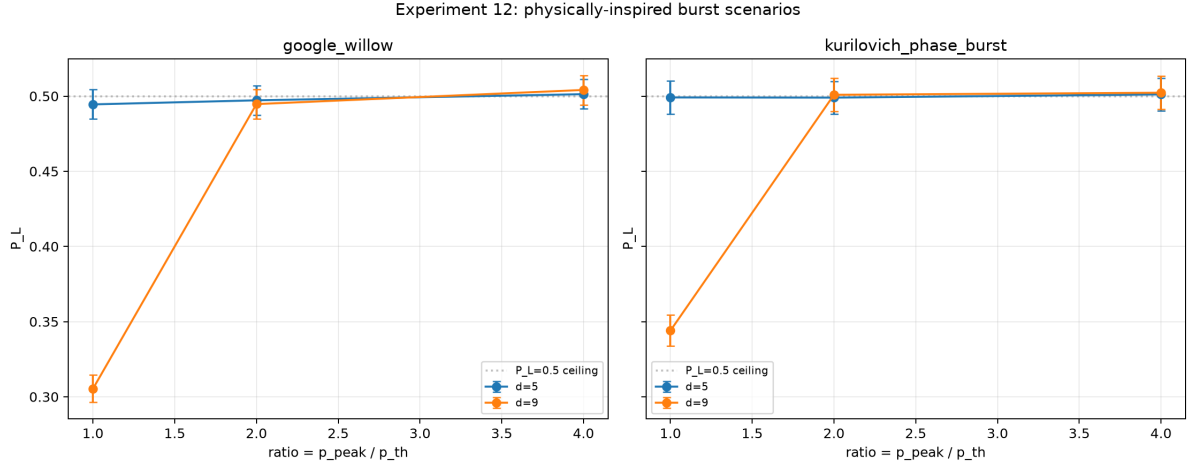


Figure 12: Logical failure probability versus strength ratio for both scenarios, at the two tested distances. The dotted line marks the saturation ceiling.

substantially more than temporal duration, and an orientation effect not present in our original design turned out to rival or exceed the size effect, growing rather than shrinking with code distance, meaning that as codes scale toward the sizes needed for real fault tolerance, a factor entirely invisible to our original hypothesis becomes more, not less, important. We further note that reading the two most relevant prior models in full, rather than at the abstract level (Tan et al., 2026; Sriram et al., 2025), twice materially changed this project’s plan, revealing both the exact textual basis for our novelty claim and a result precise enough to act on only once read completely. We record this as a methodological lesson: for a literature review to correctly determine a project’s positioning, the most load-bearing papers must be read in full.

Several limitations bound the claims made above. The low-dimensional formula we fit does not include an orientation term, and its temporal exponent was fit independently of the spatial and strength exponents, since the temporal-scaling experiment used an inconsistent round-count convention relative to the others, preventing a single joint fit with the data currently collected; a fully unified description is therefore established only along the spatial and strength axes. Converting real hardware burst durations into simulation rounds relies on an approximate, independently unverified conversion factor that could plausibly be off by a factor of a few, so the final experiment’s results should be read as order-of-magnitude rather than precise. The interpretable hint that the fitted radius exponent suggests burst area rather than linear radius as the more fundamental variable was not directly tested, due to time constraints. And the intermediate points of the angle sweep are confounded by the discrete simulation lattice not preserving exact qubit count across all rotation angles, so only the perpendicular and parallel endpoints constitute a fully controlled comparison.

6 Future Work

Our literature review documents a tension in the most recent work on burst-type noise: single, whole-code bursts have been found survivable if brief (Tan et al., 2026), while recurring whole-code bursts, observed over a sufficiently long computation, have been found to cause a two-dimensional surface code to lose its threshold entirely regardless of background noise, speculatively connected to Google’s own unexplained observation (Sriram et al., 2025). Neither tests a spatially localized, finite-radius burst. Our final experiment’s result, that a single such event, at real timescales, is essentially always catastrophic if it occurs during an active computation, makes

the frequency question urgent: if catastrophic single events are effectively guaranteed once they occur, everything now depends on how often “occur” happens, and on whether the finite spatial extent explored throughout this paper changes the qualitative conclusion reached above for the whole-code case (Sriram et al., 2025).

We propose testing whether a finite spatial burst radius, below some critical value dependent on the recurrence rate, restores a genuine asymptotic threshold for a two-dimensional surface code under bursts recurring at that rate, where a whole-code-covering burst of the same strength does not. This requires simulation infrastructure materially different from anything used above: bursts must be triggered stochastically, round by round, over memory experiments far longer than those used in this paper, whose lengths scale with code distance and typically span only five to 24 rounds; detecting whether a genuine threshold exists, rather than a single failure-probability estimate, requires sampling multiple code sizes and locating a finite-size crossing point, or fitting a two-parameter crossover scaling function analogous to one proposed elsewhere (Sriram et al., 2025); and, per the warning discussed above (Hastings et al., 2014), spatial connectivity of the burst-weakened region must be checked separately from actual code failure rather than assumed to coincide with it.

We further recommend, given the orientation finding above, that this design not restrict itself to isotropic disk bursts by default: an anisotropic burst, oriented unfavorably relative to the logical operator, may plausibly require a substantially smaller radius to be equally dangerous, and Google’s own real event is itself anisotropic. A concrete two-stage plan follows naturally: first reproduce the whole-code result discussed above (Sriram et al., 2025) within our own pipeline as a calibration step, confirming a long-horizon simulation architecture is correct before introducing the new radius variable; then introduce finite radius and repeat the genuine-threshold detection procedure across a range of radii, searching for the critical value at which the qualitative conclusion changes.

This is, by a wide margin, the highest-engineering-cost and highest-uncertainty direction identified in this report. We regard it as a well-specified, credible extension rather than a vague aspiration, precisely because the literature review and results above together identify exactly what infrastructure it requires and exactly what question it would answer, but we note explicitly that the results reported above already constitute a complete, self-contained contribution independent of whether this extension is pursued.

7 Conclusion

We opened by asking a simple question: when a cluster of qubits on a quantum chip briefly becomes noisier than the rest, for reasons that are not always known, can we predict how dangerous this is just from its size, duration, and strength relative to the error-correcting code protecting against it, without needing to know what caused it? Across thirteen controlled experiments on a surface-code quantum memory, the answer is mostly yes, in a specific and incomplete way. Relative spatial scale, how big the affected region is compared to the code and not its absolute size, is a strong and, via three independent lines of evidence including a genuine circuit-level noise model, robust predictor of the damage a burst causes, closing a gap the two most relevant prior models of burst-type noise each explicitly leave open. That picture is incomplete in a specific, structured, and now twice-validated way: which direction a burst points, relative to the code’s own logical structure, changes its danger by more than an order of magnitude in the most extreme case, and no prior model of burst damage, including our own initial one, accounts for it. Testing the resulting framework against two independently-reported real hardware events showed that, at real timescales, a single burst is essentially always catastrophic once it occurs. The question this leaves open is therefore not how bad a single burst

is, which this paper answers, but how often one occurs relative to the length of a computation, which it does not; that is the question we specify as this work’s direct continuation above.

Code and Data Availability

All code, experiment scripts, and raw results are available at <https://github.com/AlanPai777/qec-burst-scaling>. The repository contains the burst-injection and analysis library (`qecburst/`), one script per experiment (`experiments/`), and the timestamped CSV files and figures produced by each run (`results/`). Simulations were run with Stim 1.16.0 and PyMatching 2.4.0.

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